Nested While Loop in Python

Understanding Concept & Application

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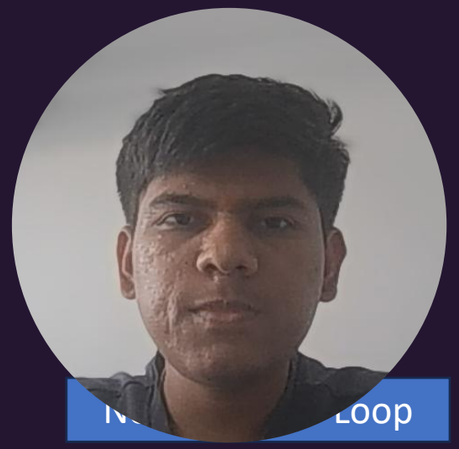
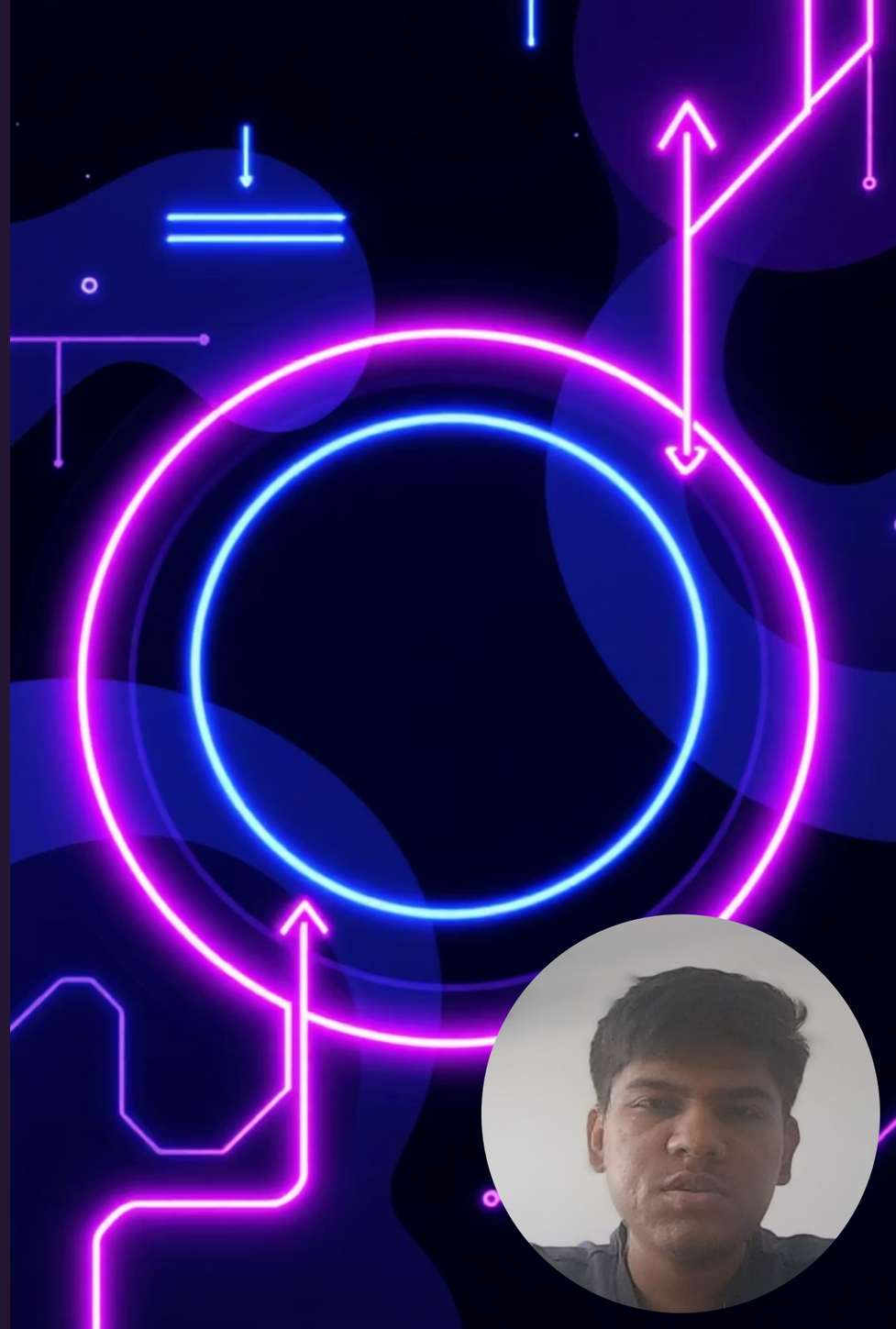
Course:Essentials Of Data Science



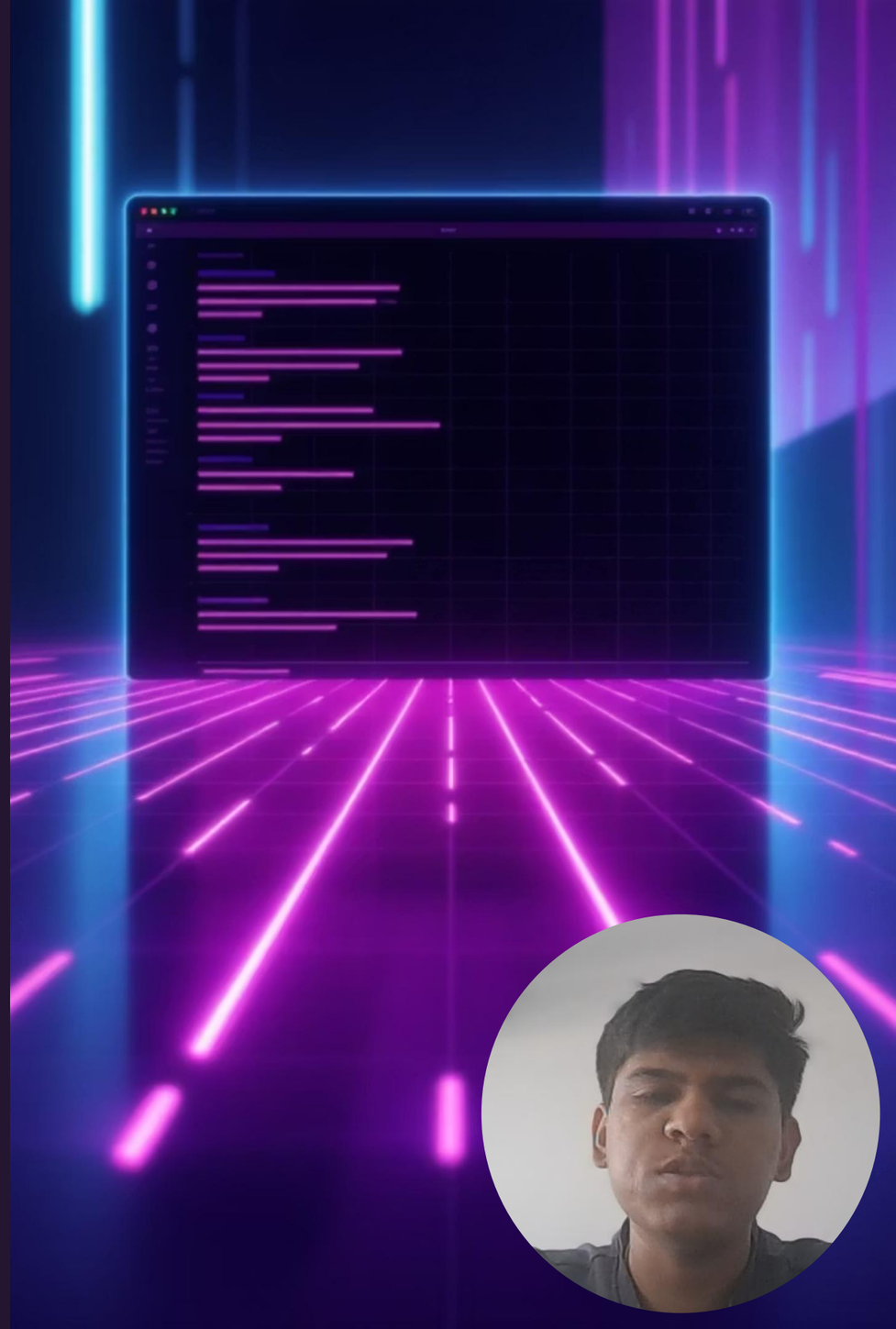
Introduction

- A loop is used to repeat a block of code.
- A **nested while loop** means a loop inside another loop.
- It is useful when working with **multi-level data** like rows and columns.
- Common in real-world problems like:
  - Pattern printing
  - Tables
  - Matrix operations



Problem Statement

- Many problems require repeating tasks **within another repetition**.
- Example:
  - Printing a grid (rows × columns)
  - Generating patterns (*, numbers)
- A single loop is not sufficient.
- Need a structured approach → **Nested loops**



Concept

Outer loop

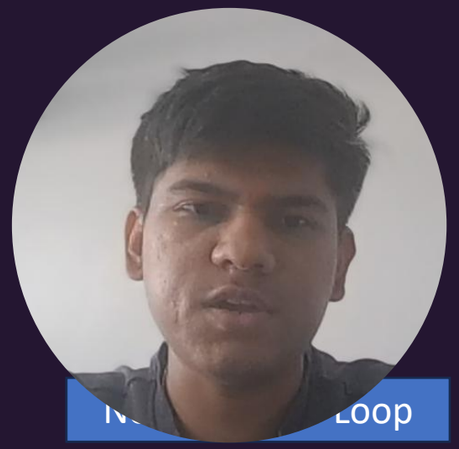
controls main iterations

Inner loop

runs completely for each outer iteration

Flow:

- Start outer loop
- Run inner loop fully
- Repeat until outer loop ends



Algorithm

01

Initialize outer loop variable

02

Start outer loop

03

Initialize inner loop variable

04

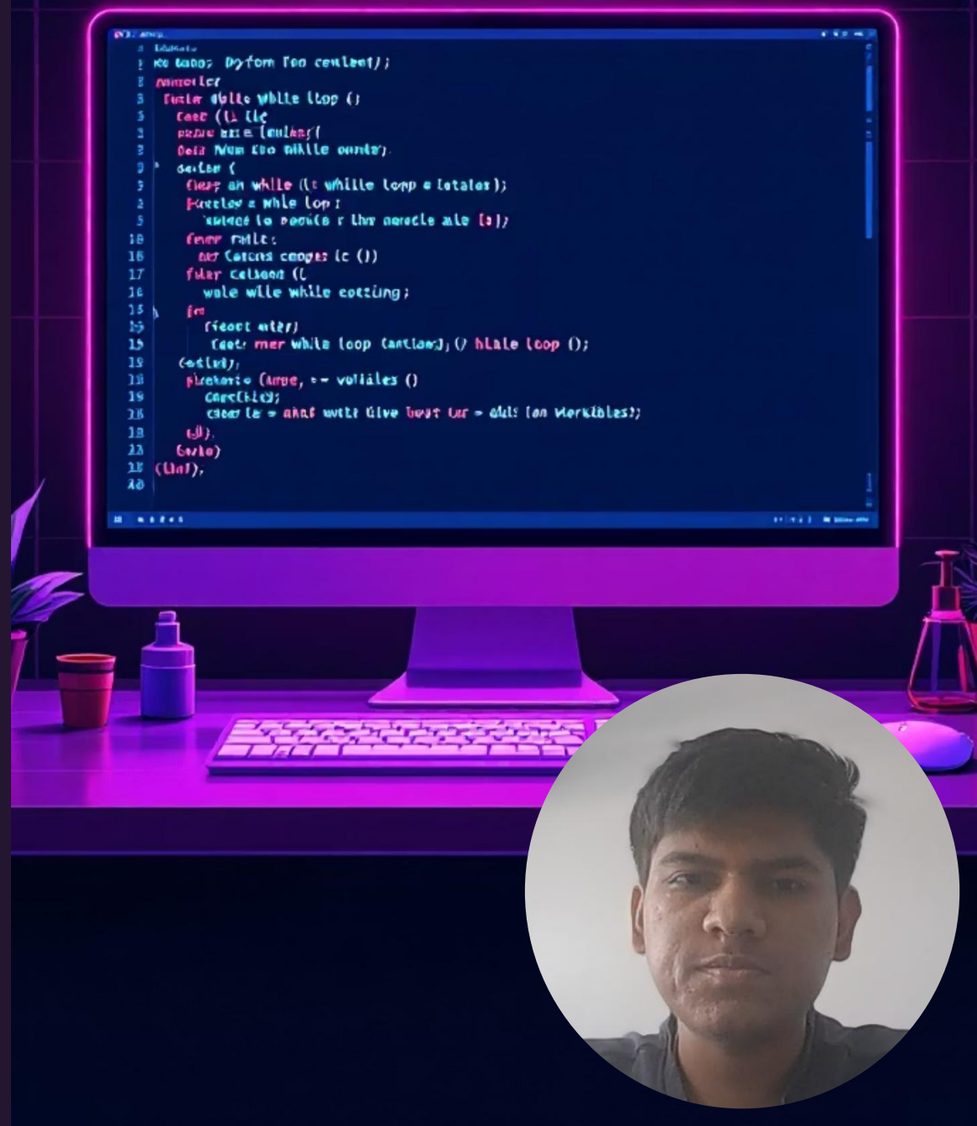
Run inner loop until condition fails

05

Move to next iteration of outer loop

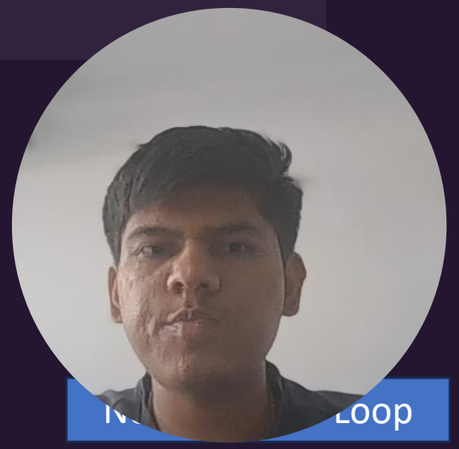
Code Example

```
i = 1
while i <= 3:
    j = 1
    while j <= 2:
        print("i =", i, "j =", j)
        j += 1
    i += 1
```



Output

```
i = 1 j = 1
i = 1 j = 2
i = 2 j = 1
i = 2 j = 2
i = 3 j = 1
i = 3 j = 2
```



Advantages

- Helps solve **multi-dimensional problems**
- Improves logical thinking
- Useful in:
  - Games
  - Data structures
  - Matrices